Jimmy Zhan
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Calculus

Feynman Integration Trick explained with an example

Consider the following integral:

$$\int_0^\infty \frac{e^{-x^2}\sin(x^2)}{x^2}dx \tag{1}$$

Parametrisation

Feynman's trick works as follows:

Denote the integral to be some function I. We need to define an integral function I of some parameter $I(a)$. In this example, we place the parameter as part of the argument of the sine function.

$$I(a) = \int_0^\infty \frac{e^{-x^2}\sin(ax^2)}{x^2}dx \tag{2}$$

It is a sensible choice because if we take the partial derivative of $\sin(ax^2)$ with respect to a, we would get $x^2\cos(ax^2)$, thereby simplifying the integral.

Differentiating $I(a)$ with respect to a gives

$$\frac{d}{da}I(a) = \frac{d}{da}\int_0^\infty \frac{e^{-x^2}\sin(ax^2)}{x^2}dx \tag{3}$$

Let's check if we can swap the integration and differentiation operators by implementing the Leibniz Rule and Dominated Convergence Theorem.

Convergence

Theorem 1. *Let*

$$I(a) = \int_0^\infty f(x,a)dx \tag{4}$$

.

If:

1. For each fixed a, the function $x \mapsto f(x,a)$ is integrable on $(0,\infty)$.

2. The partial derivative $\partial_a f(x,a)$ exists for all $x > 0$ and all a in an open interval $U \subset \mathbb{R}$.

3. There exists a function $g \in L^1(0,\infty)$, independent of a, such that

$$|\partial_a f(x,a)| \leqslant g(x) \tag{5}$$

for all $x > 0$ and all $a \in U$.

Then I is differentiable on U, and

$$\frac{d}{da}I(a) = \int_0^\infty f(x,a)dx \tag{6}$$

Rule 1

Since $\sin t \sim t$ $(t \to 0)$, we have $\frac{\sin(ax^2)}{x^2} \sim \frac{ax^2}{x^2} = a$. Therefore, $f(x,a) \to a$ as $x \to 0$. The function is integrable at $x = 0$.

Also, $e^{-x^2}\cos(ax^2)$ is continuous and bounded at $x = 0$.

For each fixed a, the function $x \mapsto f(x,a)$ is integrable on $x = 0$. It is obvious to see that the function is integrable on $(0,\infty)$.

Rule 2

$$\partial_a f(x,a) = e^{-x^2}\cos(ax^2) \tag{7}$$

Rule 3

Denote

$$f(x,a) = \frac{e^{-x^2}\sin(ax^2)}{x^2} \tag{8}$$

We have

$$\frac{\partial}{\partial a} f(x,a) = e^{-x^2}\cos(ax^2) \tag{9}$$

Therefore

$$|\partial_a f(x,a)| = |e^{-x^2}\cos(ax^2)| \leqslant e^{-x^2} \tag{10}$$

Also

$$\int_0^\infty e^{-x^2} dx < \infty \tag{11}$$

Therefore we may take $g(x) = e^{-x^2}$.

Feynman's Trick - Differentiation part

From the chapter **Convergence**, we may conclude that

$$\frac{d}{da}\int_0^\infty \frac{e^{-x^2}\sin(ax^2)}{x^2} dx = \int_0^\infty \frac{\partial}{\partial a}\frac{e^{-x^2}\sin(ax^2)}{x^2} dx \tag{12}$$

Therefore we have

$$\begin{aligned} I'(a) &= \int_0^\infty \frac{\partial}{\partial a}\frac{e^{-x^2}\sin(ax^2)}{x^2} dx \\ &= \int_0^\infty e^{-x^2}\cos(ax^2) dx \end{aligned} \tag{13}$$

From Euler's Formula, $e^{ix} = \cos x + i\sin x$. Therefore, $\Re(e^{iax^2}) = \cos(ax^2)$, which means that

$$\begin{aligned} I'(a) &= \Re(\int_0^\infty e^{-x^2} e^{iax^2} dx) \\ &= \Re(\int_0^\infty e^{-x^2(1-ia)} dx) \end{aligned} \tag{14}$$

This integral is just an example of a generalised Gaussian integral.

Lemma 2. *Gaussian integral*
The following equality holds if $s \in \mathbb{C}$ and $\Re(s) > 0$.

$$\int_0^\infty e^{-sx^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{s}} \tag{15}$$

Therefore

$$
\begin{aligned}
I'(a) &= \frac{1}{2}\sqrt{\pi}\Re(\frac{1}{\sqrt{1-ia}}) \\
&= \frac{\sqrt{\pi}}{2}\Re((1-ia)^{-1/2})
\end{aligned}
\tag{16}
$$

Now that we have the derivative of I with respect to a completely in terms of the parameter a alone.

Feynman's Trick - Integration part

$$
\begin{aligned}
\int I'(a)\ da &= \frac{\sqrt{\pi}}{2}\ \Re(\int(1-ia)^{-1/2}da) \\
I(a) &= \frac{\sqrt{\pi}}{2}\ \Re\left(\frac{(1-ia)^{1/2}}{(\frac{1}{2})(-i)}\right) + C \\
I(a) &= \sqrt{\pi}\ \Re\left(\frac{\sqrt{1-ia}}{-i}\right) + C \\
&= \sqrt{\pi}\ \Re\left(i\sqrt{1-ia}\right) + C
\end{aligned}
\tag{17}
$$

Recall that $I(a)$ is defined as

$$
I(a) = \int_0^\infty \frac{e^{-x^2}\sin(ax^2)}{x^2}dx \tag{18}
$$

Therefore

$$
I(0) = 0 \tag{19}
$$

Combine this with (17) we have

$$
\begin{aligned}
0 &= \sqrt{\pi}\ \Re\left(i\right) + C \\
C &= 0
\end{aligned}
\tag{20}
$$

Therefore, we can focus on the target integral.

$$
I = I(1) = \sqrt{\pi}\ \Re\left(i\sqrt{1-i}\right) \tag{21}
$$

We now use the polar representation of complex numbers.
Let $z = 1-i$. We have $|z| = \sqrt{2}$ and $\arg(z) = \tan^{-1}(\frac{-1}{1}) = -\frac{\pi}{4}$.
Therefore

$$
\begin{aligned}
z &= 1 - i = \sqrt{2}e^{\frac{-i\pi}{4}} \\
\sqrt{z} = \sqrt{1-i} &= \sqrt{\sqrt{2}e^{\frac{-i\pi}{4}}} \\
\sqrt{1-i} &= \sqrt{\sqrt{2}}e^{\frac{-i\pi}{8}} \\
i\sqrt{1-i} &= i\sqrt{\sqrt{2}}\left(\cos\frac{\pi}{8} - i\sin\frac{\pi}{8}\right) \\
i\sqrt{1-i} &= \sqrt{\sqrt{2}}\left(i\cos\frac{\pi}{8} + \sin\frac{\pi}{8}\right) \\
\Re\left(i\sqrt{1-i}\right) &= \Re\left(\sqrt{\sqrt{2}}\left(i\cos\frac{\pi}{8} + \sin\frac{\pi}{8}\right)\right) \\
&= \sqrt{\sqrt{2}}\sin\frac{\pi}{8}
\end{aligned} \tag{22}
$$

Back to (21) we have

$$
\int_0^\infty \frac{e^{-x^2}\sin(x^2)}{x^2}dx = \sqrt{\pi\sqrt{2}}\sin\frac{\pi}{8} \tag{23}
$$